

Sawaira Mumtaz

AI & Software Engineer | LLM, RAG & Chatbot Systems

📍 Rawalpindi, Pakistan (Open to Onsite & Remote)
☎ +923155069089
✉ sawairamumtaz369@gmail.com
🔗 Portfolio: <https://sawaira-mumtaz786.github.io/Portfoliowebsite/>
💻 GitHub: github.com/Sawaira-Mumtaz786
🔗 LinkedIn: linkedin.com/in/sawaira-mumtaz-3b77972b1

SUMMARY

Computer Science graduate specializing in production ready LLM applications, RAG pipelines, chatbot systems, and scalable full-stack software. Experienced across AI engineering, data science, mobile development, and full-stack engineering through hands-on work with 4 companies. Passionate about building AI systems that ship to production and solve real-world problems. Actively seeking entry-level AI & Software Engineering roles open to onsite and remote positions worldwide, as well as freelance projects.

EDUCATION

- Matriculation (Science) (2018)
Government Girls High School GareebAbad (Rawalpindi)
 - Intermediate (ICS-Physics) (2021)
Degree College for Women Dhoke Munshi (Rawalpindi)
 - Bachelors in Computer Science (8th semester continue...)
Virtual University of Pakistan (Saddar Campus)
-

Final Year Project : AI-Powered Clothing & Inventory Management System originally developed as a desktop based application in Python + Tkinter with SQLite (DBMS), featuring separate Admin and Staff modules for inventory control, stock tracking, and billing. Later extended with AI capabilities by integrating Google Gemini 2.5 Flash for vision based auto tagging (upload a photo AI extracts item name, category, season, size, and color), then rebuilt as a Streamlit web application deployed publicly on GitHub.

Specialization: Production ready LLM applications, RAG pipelines, chatbot systems, and scalable full stack software.

PROJECTS

ADVANCED AI SYSTEMS: LLM EVAL & MULTI-AGENT

Python · LangGraph · Streamlit · Groq API · Plotly

- Built an LLM evaluation dashboard using LLM as judge methodology to score outputs on Faithfulness

(hallucination detection) and Answer Relevance cutting manual review time by 70%.

- Designed a LangGraph multi-agent state machine where Agent A (Structurer) converts messy text into clean JSON and Agent B (Validator) applies business rules, with Human-in-the-Loop approval checkpoints.

NOTIFICATIONS & AI INCIDENT ENGINE

React · TypeScript · NestJS · JWT · Google Gemini AI

- Full-stack enterprise application with modular NestJS backend (Controllers, Services, DTOs, Guards).
- Implemented JWT auth + bcrypt (10 rounds) to replace MD5 and cookie-based sessions reducing incident resolution time by 40%.
- Integrated Google Gemini AI for auto-triage, urgency scoring (1–10), and RCA playbooks.

AI CLOTHING & INVENTORY MANAGEMENT SYSTEM (Final Year Project)

Python · Tkinter · SQLite · Streamlit · Google Gemini 2.5 Flash

- Desktop-based inventory system with Admin and Staff modules for stock tracking, billing, and profit analytics.
- Extended with AI vision-based auto-tagging via Gemini 2.5 Flash reducing manual entry time by over 80%.
- Rebuilt as a Streamlit web app and deployed publicly.

MGT501 RAG AI CHATBOT

Python · Flask · Google text-embedding-004 · Gemini

- RAG chatbot that converts 45+ course handouts into vector embeddings and uses cosine similarity search for Top K retrieval.
- Enforces strict out-of-domain refusal to prevent hallucinated answers on non-course queries.

POSTS FUSION STUDIO — AI CONTENT GENERATOR

Node.js · Express · JavaScript · Server-Sent Events (SSE)

- Real-time AI content streaming app with word-by-word rendering via SSE.
- Two generation modes (Streaming + Full Post) with easy swapping between OpenAI, Groq, and OpenRouter cuts content creation time from 30 min to under 3 min per post.

COLLEGE ALERT APP | E-COMMERCE MARKETPLACE APP

DIGITAL ART GALLERY | TIME ZONE CONVERTER

React Native · Expo · JavaScript

- Suite of 4 cross-platform mobile apps: event management with push notifications, multi-screen e-commerce marketplace, React Native Web art gallery, and Luxon-powered timezone converter covering 9 major global zones.

TECHNICAL COMPUTER SKILLS

AI & MACHINE LEARNING

- LLM Integration (OpenAI, Gemini, Groq, OpenRouter, Ollama)
- RAG Systems, Vector Databases (pgvector, Chroma)
- LangChain, LangGraph, HuggingFace
- Prompt Engineering, Semantic Search, AI Agents
- Machine Learning, Deep Learning, Model Testing

- Scikit-learn, Pandas, NumPy, K-Means, SVD

PROGRAMMING LANGUAGES

- Python, JavaScript, TypeScript, C, C++

FRAMEWORKS & LIBRARIES

- React, React Native, Node.js, Express.js, NestJS
- Flask, Streamlit, Tkinter, Tailwind CSS, Bootstrap
- TensorFlow.js, Mongoose

DATABASES & AUTHENTICATION

- MongoDB, PostgreSQL, SQLite, Supabase
- JWT Authentication, bcrypt, Secure User Auth, CRUD

APIs & INTEGRATIONS

- REST APIs, Flask APIs, Server-Sent Events (SSE)
- Google Gemini API, OpenAI API, Groq API, OpenRouter

DEVOPS & TOOLS

- Git, GitHub, GitLab CI/CD, Docker, Nginx, Linux
- Render, Railway, Vercel, Google Colab, Jupyter
- Agile / Scrum

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEERING (4 Mar 2026 – 4 Jun 2026)

GO Viral Communication, ISB NASTP — Islamabad, Pakistan (Onsite)

- Developed a full-stack Notifications System using React TypeScript (frontend), NestJS (backend), MongoDB, and RESTful APIs.
- Implemented secure authentication using JWT token-based auth and bcrypt password hashing for user data protection.
- Built a comprehensive backend with CRUD operations, data validation, and RESTful architecture.
- Gained practical experience in C, Python Flask APIs, Git/GitHub, Linux, Docker, and Nginx

web server administration.

AI ENGINEERING AI & ML DEPARTMENT (1 Jun 2026 – 28 Aug 2026)

Post Fusion (Pro Fusion AI) — Rawalpindi, Pakistan (Onsite)

- Developed a LinkedIn Post Generator using Python, LangChain, and Ollama, implementing prompt engineering and AI quality scoring.
 - Designed and deployed an AI-powered content generation studio integrating multiple LLM APIs (OpenAI, Groq, OpenRouter) with real-time streaming via Server-Sent Events (SSE).
 - Built a full-stack streaming content platform using React + Node.js/Express, vector databases, and multi-LLM routing.
 - Mastered advanced concepts: Machine Learning, Deep Learning, Prompt Engineering, LLM Applications, AI Workflows, and AI Agents.
 - Implemented version control and CI/CD workflows using Git and GitHub under professional mentorship.
-

DATA SCIENCE — SELF LEARNING (Jul 2026 – Sep 2026)

Self-Directed Study & Personal Projects

- Built a collaborative filtering recommendation system (SVD algorithm) using the Surprise library, achieving industry-standard RMSE scores for rating predictions.
 - Applied K-Means clustering (Scikit-learn) to segment retail customers into 5 distinct personas, enabling targeted marketing strategies.
 - Performed data cleaning, visualization, and statistical analysis using Python, Pandas, and NumPy for business intelligence insights.
-

MOBILE DEVELOPMENT — SELF LEARNING (Jul 2026 – Sep 2026)

Self-Directed Study & Personal Projects

- Built cross-platform mobile apps using Flutter and React Native with responsive UIs.
- Integrated REST APIs and backend services for seamless data flow.
- Implemented modern state management and navigation patterns across multiple screens.

- Delivered fully functional mobile prototypes aligned with industry design standards.
-

CERTIFICATIONS

- Introduction to cyber security
 - Certified Ethical hacking
 - Freelancing
 - Introduction to C++
 - AI content generator
 - Certificate Information Technology
 - Web penetration testing
 - Advance Web Application development
 - Wordpress
 - Data analytics & business intelligence
 - Cloud computing
 - UI/UX & WEBFLOW
-

PERSONAL SKILLS:

- The first major feature is ambitions. I am a very ambitious person, responsible, reliable, flexible, enthusiastic, and dynamic.
 - Quick-learning, motivated with good communication abilities, fast integration in a group / Team, result-oriented. Able to work independently in a stressed environment.
 - Excellent qualities such as teamwork, leadership, negotiation, analysis, and communication.
 - Strong skills in the areas of project management, people management, systems design, project preparation and implementation, technical support, commissioning, training, troubleshooting.
-

SOFT SKILLS

- Team Work & Collaboration
 - Communication & Interpersonal Skills
 - Leadership & Team Lead Experience
 - Time Management & Task Prioritization
 - Problem-Solving & Issue Resolution
 - Result-Oriented & Self-Motivated
 - Quick Learner, Adaptable to New Technologies
 - Presentation & Public Speaking
-

LANGUAGES

- English — Professional Working Proficiency
 - Urdu — Native
-